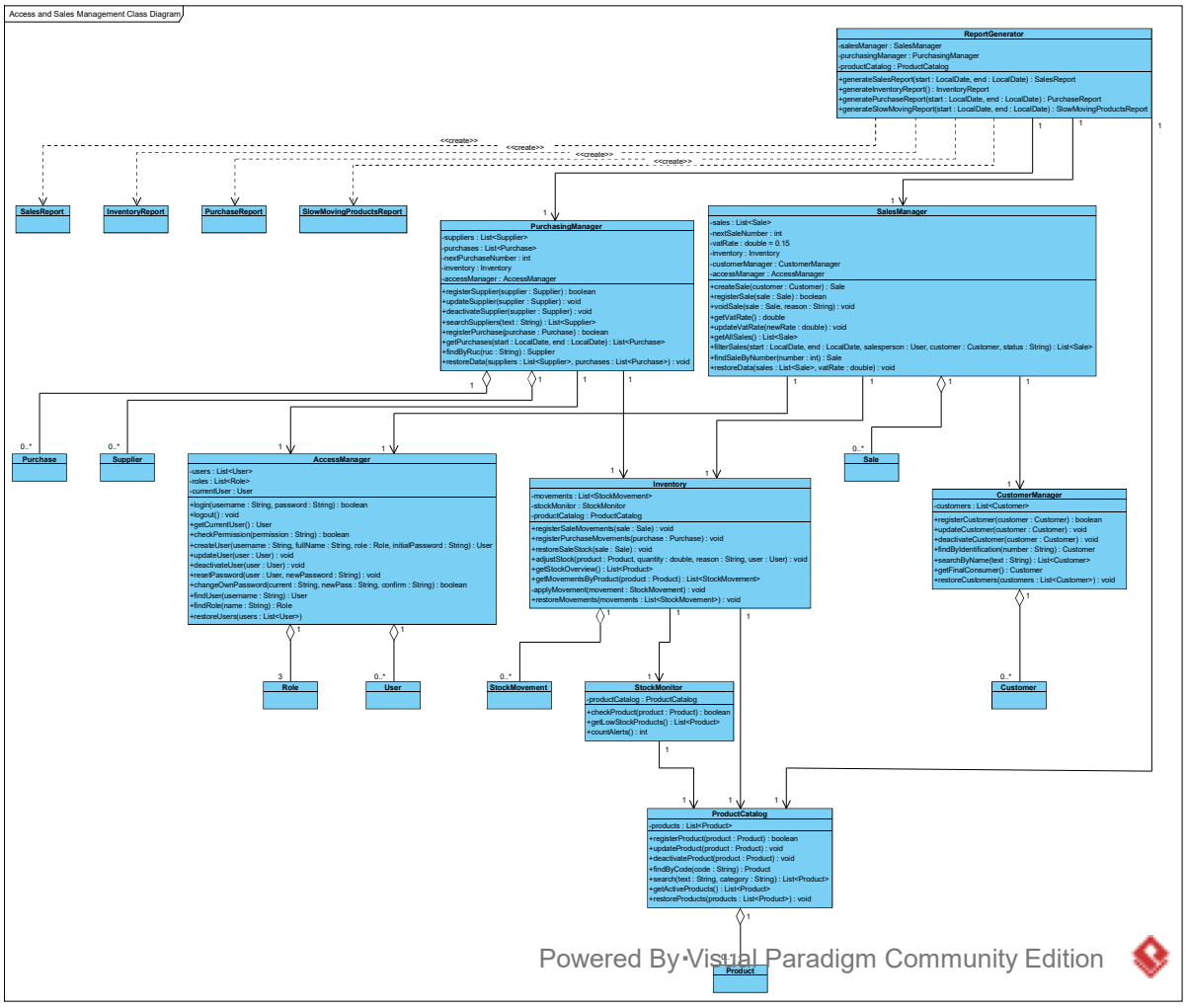

Access and Sales Management Class Diagram
ReportGenerator
-salesManager : SalesManager
-purchasingManager : PurchasingManager
-productCatalog : ProductCatalog
+generateSalesReport(start : LocalDate, end : LocalDate) : SalesReport
+generateInventoryReport() : InventoryReport
+generatePurchaseReport(start : LocalDate, end : LocalDate) : PurchaseReport
+generateSlowMovingReport(start : LocalDate, end : LocalDate) : SlowMovingProductsReport
<<create>>
SalesReport
InventoryReport
PurchaseReport
SlowMovingProductsReport
PurchasingManager
-suppliers : List<Supplier>
-purchases : List<Purchase>
-nextPurchaseNumber : int
-inventory : Inventory
-accessManager : AccessManager
+registerSupplier(supplier : Supplier) : boolean
+updateSupplier(supplier : Supplier) : void
+deactivateSupplier(supplier : Supplier) : void
+searchSuppliers(text : String) : List<Supplier>
+registerPurchase(purchase : Purchase) : boolean
+getPurchases(start : LocalDate, end : LocalDate) : List<Purchase>
+findByRuc(ruc : String) : Supplier
+restoreData(suppliers : List<Supplier>, purchases : List<Purchase>) : void
SalesManager
-sales : List<Sale>
-nextSaleNumber : int
-vatRate : double = 0.15
-inventory : Inventory
-customerManager : CustomerManager
-accessManager : AccessManager
+createSale(customer : Customer) : Sale
+registerSale(sale : Sale) : boolean
+voidSale(sale : Sale, reason : String) : void
+getVatRate() : double
+updateVatRate(newRate : double) : void
+getAllSales() : List<Sale>
+filterSales(start : LocalDate, end : LocalDate, salesperson : User, customer : Customer, status : String) : List<Sale>
+findSaleByNumber(number : int) : Sale
+restoreData(sales : List<Sale>, vatRate : double) : void
Purchase
Supplier
AccessManager
-users : List<User>
-roles : List<Role>
-currentUser : User
+login(username : String, password : String) : boolean
+logout() : void
+getCurrentUser() : User
+checkPermission(permission : String) : boolean
+createUser(username : String, fullName : String, role : Role, initialPassword : String) : User
+updateUser(user : User) : void
+deactivateUser(user : User) : void
+resetPassword(user : User, newPassword : String) : void
+changeOwnPassword(current : String, newPass : String, confirm : String) : boolean
+findUser(username : String) : User
+findRole(name : String) : Role
+restoreUsers(users : List<User>)
Inventory
-movements : List<StockMovement>
-stockMonitor : StockMonitor
-productCatalog : ProductCatalog
+registerSaleMovements(sale : Sale) : void
+registerPurchaseMovements(purchase : Purchase) : void
+restoreSaleStock(sale : Sale) : void
+adjustStock(product : Product, quantity : double, reason : String, user : User) : void
+getStockOverview() : List<Product>
+getMovementsByProduct(product : Product) : List<StockMovement>
-applyMovement(movement : StockMovement) : void
+restoreMovements(movements : List<StockMovement>) : void
Sale
CustomerManager
-customers : List<Customer>
+registerCustomer(customer : Customer) : boolean
+updateCustomer(customer : Customer) : void
+deactivateCustomer(customer : Customer) : void
+findByIdentification(number : String) : Customer
+searchByName(text : String) : List<Customer>
+getFinalConsumer() : Customer
+restoreCustomers(customers : List<Customer>) : void
Role
User
StockMovement
StockMonitor
-productCatalog : ProductCatalog
+checkProduct(product : Product) : boolean
+getLowStockProducts() : List<Product>
+countAlerts() : int
Customer
ProductCatalog
-products : List<Product>
+registerProduct(product : Product) : boolean
+updateProduct(product : Product) : void
+deactivateProduct(product : Product) : void
+findByCode(code : String) : Product
+search(text : String, category : String) : List<Product>
+getActiveProducts() : List<Product>
+restoreProducts(products : List<Product>) : void
Product
Powered By Visual Paradigm Community Edition